

Recitation #5

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Convexity and Gradient Descent

1. Determine whether the function $f(x, y) = 2x^2 + 3y^2 - xy$ is convex by calculating the Hessian matrix.

Solution. First, compute the first partial derivatives:

$$\frac{\partial f}{\partial x} = 4x - y, \quad \frac{\partial f}{\partial y} = 4y - x.$$

Differentiating again gives

$$\frac{\partial^2 f}{\partial x^2} = 4, \quad \frac{\partial^2 f}{\partial y^2} = 4, \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = -1.$$

Thus the Hessian is

$$H(f) = \begin{pmatrix} 4 & -1 \\ -1 & 4 \end{pmatrix}.$$

To find its eigenvalues, solve $\det(H(f) - \lambda I) = 0$:

$$\begin{aligned} \det \begin{pmatrix} 4 - \lambda & -1 \\ -1 & 4 - \lambda \end{pmatrix} &= (4 - \lambda)^2 - 1 \\ &= \lambda^2 - 8\lambda + 15 \\ &= (\lambda - 3)(\lambda - 5) = 0. \end{aligned}$$

Therefore the eigenvalues are $\lambda = 3$ and $\lambda = 5$. Both are positive, so the Hessian is positive definite. Hence f is strictly convex, and therefore convex.

2. Show that the L2 regularization term $f(\beta) = \|\beta\|_2^2$ is convex. Use the second derivative test for this function.

Solution. Write $f(\beta) = \sum_{j=1}^p \beta_j^2$. Then

$$\frac{\partial^2 f}{\partial \beta_j^2} = 2, \quad \frac{\partial^2 f}{\partial \beta_j \partial \beta_k} = 0 \quad (j \neq k).$$

Thus $\nabla^2 f(\beta) = 2I$, which is positive definite. So f is strictly convex.

3. For the linear regression model $y = X\beta$ with an L2 regularization term $\|\beta\|_2^2$, the objective function becomes $L(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|_2^2$, where $\lambda > 0$ is a regularization parameter. Compute the gradient of $L(\beta)$ with respect to β .

Solution. Expanding the objective gives

$$L(\beta) = y^T y - 2\beta^T X^T y + \beta^T X^T X \beta + \lambda \beta^T \beta.$$

Taking the gradient,

$$\nabla_{\beta} L(\beta) = -2X^T y + 2X^T X \beta + 2\lambda \beta = 2X^T (X\beta - y) + 2\lambda \beta.$$

4. Consider the linear regression model $y = X\beta$, where $X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}$ and $y = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$. Starting with $\beta_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, perform one iteration of gradient descent with learning rate $\alpha = 0.01$.

Solution. The gradient of the squared-error loss is

$$\nabla_{\beta} L(\beta) = 2X^T (X\beta - y).$$

At $\beta_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, the predictions are $X\beta_0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$. Thus

$$\nabla_{\beta} L(\beta_0) = 2 \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} -1 \\ -2 \\ -2 \end{pmatrix} = 2 \begin{pmatrix} -1 - 2 - 2 \\ -1 - 4 - 6 \end{pmatrix} = \begin{pmatrix} -10 \\ -22 \end{pmatrix}.$$

Applying the gradient descent update with $\alpha = 0.01$,

$$\begin{aligned} \beta_1 &= \beta_0 - \alpha \nabla_{\beta} L(\beta_0) \\ &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} - 0.01 \begin{pmatrix} -10 \\ -22 \end{pmatrix} \\ &= \boxed{\begin{pmatrix} 0.10 \\ 0.22 \end{pmatrix}}. \end{aligned}$$

Probability

1. You roll a fair six-sided die.
 - (a) What is the sample space?
 - (b) Define the event “rolling an even number.” What is its probability?

- (c) Define the random variable $X = (\text{outcome mod } 2)$. Give the distribution of X .

Solution. (a) $\Omega = \{1, 2, 3, 4, 5, 6\}$. (b) $E = \{2, 4, 6\}$, so $P(E) = 3/6 = 1/2$. (c) X takes values $\{0, 1\}$. $P(X = 0) = P(\text{even}) = 1/2$, $P(X = 1) = 1/2$.

2. A fair coin is flipped 3 times.

- (a) What is the probability of exactly two heads?
 (b) What is the probability of at least one head?
 (c) If the coin were biased with $P(H) = 0.7$, what changes?

Solution. (a) $\binom{3}{2}(1/2)^3 = 3/8$. (b) $1 - P(0 \text{ heads}) = 1 - (1/2)^3 = 7/8$. (c) For $p = 0.7$, let $E_{i,j}$ be the event “toss i and toss j are the only heads.” Then $P(E_{i,j}) = p^2(1-p)$. So $P(\text{exactly two heads}) = P(E_{1,2}) + P(E_{2,3}) + P(E_{1,3}) = 3p^2(1-p)$. Likewise $P(\text{at least one head}) = 1 - P(\text{no heads}) = 1 - (1-p)^3 = 1 - 0.3^3 = 0.973$.

3. Two fair dice are rolled. Let X =value of die 1, Y =value of die 2.

- (a) What is $P(X = 3, Y = 5)$?
 (b) What is $P(X = 3 \mid X + Y = 8)$?
 (c) Are X and Y independent?

Solution. (a) Each outcome equally likely with prob $1/6$, so $1/36$. (b) Outcomes with sum 8: $(2,6), (3,5), (4,4), (5,3), (6,2)$. 5 equally likely. Only one has $X = 3$. So $1/5$. (c) Independence: for any a, b , $P(X = a, Y = b) = 1/36 = P(X = a)P(Y = b)$. Yes, they are independent.